* Wrapping function

→ way to change the shape or position of an image by modifying the location of its pixels.

→ It works on Spatial Domain

→ Given an image I warping can be seen as a function w that computes the new coordinates of each pixels

$$u' = w_u(u,v) \quad , v' = w_v(u,v)$$

The transformation can be

Rotation $\begin{bmatrix} u' \\ v' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}$

Full homography: (r, t, scalings & perspective change)

$$\begin{bmatrix} u' \\ v' \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} = \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}$$

* Forward mapping

→ Starting from the i/p image coordinates, apply the warping function to obtain the o/p.

→ the output coordinates might be continuous and need to be discretized (truncated, rounded)
→ this might have two problem.

Fold → more than pixel end up same place
hole → o/p pixel dont have i/p pixel (empty).

* Backward mapping

→ starts from o/p image coordinates, use the inverse of the warping function w^{-1} to find it corresponding o/p

$$u = w^{-1}_u(u', v') \qquad v = w^{-1}_v(u', v')$$

The i/p might be continuous, Possible discretization strategies
→ truncation, N.N interpolation b/w 4 closes pixels

* Bilinear Interpolation

1. Find the 4 closest pixels surrounding (u,v)
2. Compute how far (u,v) is from the top left pixel
3. interpolate the horizontal direction:
4. Interpolate in the vertical direction b/w I_a & I_b